

Assignment 2: Sanskrit-to-English Neural Machine Translation

Natural Language Understanding

25 June 2026

1 Overview

Build a Neural Machine Translation (NMT) system for translating Sanskrit text into English. Sanskrit is a low-resource, morphologically rich language, making translation challenging.

2 Problem Statement

Given a Sanskrit sentence, generate its corresponding English translation using a custom sequence-to-sequence architecture.

3 Dataset

Parallel Sanskrit–English sentence pairs (training, dev, test sets). Files: `train_sa.csv`, `dev_sa.csv`, `test_sa.csv`, `train_en.csv`, `dev_en.csv`, `test_en.csv`. Each file contains `Source_id` and `Sentence_sa`/`Sentence_en`. `Source_id` aligns sentence pairs across files.

The dataset can be downloaded using the link.

Source_id	Sentence_sa	Source_id	Sentence_en
1	"Ctrl, S नुत्ता रक्षन्तु।"	1	Save it with Ctrl, S.
2	गुरुः छात्रान् एकवारं पाठयति ।	2	Teacher will teach the students only once.

{train/dev/test}_sa.csv {train/dev/test}_en.csv

Figure 1: Format of the Data: Sanskrit text and its corresponding English translation. (`Source_id` is the common ID between two files)

4 Task

Model Development: Build Sanskrit-to-English translation model using custom seq2seq architecture. Generate English translations for test set.

5 Required Outputs (Test Set)

File: `submission.csv` (UTF-8 encoded)

Format: Two columns - `Source_id`, `Sentence_en`. Each row contains predicted English translation for given `Source_id`.

Example:

Source_id	Sentence_en
1	The meaning of duty is explained
2	The people of the nation are united
3	Knowledge leads to liberation

6 Evaluation Metrics

1. BLEU - BLEU is computed between predicted english sentences and gold sentences from the test set. Use the default NLTK BLEU score¹ with no weights.
2. BERTScore - BERTScore is computed between predicted english sentences and gold sentences from the test set. Use BERTScore in Python via the official bert-score library, available on PyPI². Report only F1 BERTScore and use *rescale_with_baseline = True*.
3. Efficiency: Write a code to display inference time (total time taken to generate translation for the test set) and the total number of parameters in your model.

Scoring: $\text{Marks}_{\text{metric}} = \frac{\text{Student Score}}{\text{Highest Score}} \times \text{Max Marks}_{\text{metric}}$, here the metrics are BLEU and BERTScore.

For efficiency, $\text{Marks}_{\text{metric}} = \frac{\text{Minimum Score}}{\text{Student Score}} \times \text{Max Marks}_{\text{metric}}$

Final Score:

$$\begin{aligned} \text{Score} = & 0.4 \times \text{Marks}_{\text{BLEU}} \\ & + 0.3 \times \text{Marks}_{\text{BERTScore}} \\ & + 0.15 \times \text{Marks}_{\text{Efficiency}_{\text{Time}}} \\ & + 0.15 \times \text{Marks}_{\text{Efficiency}_{\text{parameters}}} \end{aligned}$$

7 Submission

ZIP Contents: (1) `submission.csv`, (2) Report (PDF), (3) Code (.ipynb only, **no .py**)

Report Sections (10-11pt font): Introduction, Architecture (encoder, decoder, attention, beam search), Training Procedure (preprocessing, tokenization, hyperparameters, optimization, regularization), Results (BLEU, BERTScore, Efficiency on dev/test, training curves), Translation Examples (5-10 examples with source, reference, prediction, error analysis), Experiments, Discussion (design choices, challenges, limitations), References. Disclose all pre-trained models used.

8 Rules

- Only the provided dataset should be used for this assignment.
- Pre-trained models are permitted, but their usage must be clearly disclosed in the report.
- All submitted work must be original and fully reproducible.
- Usage of any external APIs is strictly prohibited. The code will be checked for API calls.
- Submitted code will be evaluated for similarity. Any detected plagiarism will result in penalties.
- Each student must submit individually. Group submissions are not allowed.
- Each .ipynb file must include installation steps for all required dependencies.
- Any package or framework may be used for implementation; however, the use of PyTorch is recommended, and excessive use of external packages should be avoided to ensure faster evaluation.
- The code will be evaluated on a private dataset that will be released only at the time of evaluation.
- Final evaluation will be conducted after the submission deadline. The private test set will be released at that time, and the obtained scores will be collected for grading.
- All codes have to be uploaded on github and should be made public.

¹https://www.nltk.org/api/nltk.translate.bleu_score.html

²<https://pypi.org/project/bert-score/>

9 Deadline

The deadline for submission is 4th July (11:59 PM). Evaluation will be taken in class on 5th July where private dataset will released and you will asked to submit scores on the that dataset.

10 Marks Distribution

Component	Marks
Evaluation Metrics (80%)	80
Report (20%)	20
Total	100